From a text to word counts

One six-word text:

banana **chocolate** **banana** **fudge** **banana** **chocolate**

	banana	**chocolate**	**fudge**
Observed count	3	2	1

- A **word distribution** assigns a probability to each category.
- We ask how plausible these counts are for a particular speaker's distribution.

Next: how likely are these counts for a speaker?

Multinomial: keep the word categories separate

	banana	chocolate	fudge
Observed count	3	2	1
Speaker A's probability	0.5	0.3	0.2

$$P(3, 2, 1 \mid A) = \frac{6!}{3!\,2!\,1!}(0.5)^3(0.3)^2(0.2)^1 = 0.135.$$

- Counts add to the text length: $3 + 2 + 1 = 6$.
- Word probabilities add to one: $0.5 + 0.3 + 0.2 = 1$.
- The model counts all word orders with this same pattern of counts.

Why keep more than one word?

	banana	chocolate	fudge
Speaker A	0.5	0.3	0.2
Speaker B	0.5	0.1	0.4

- Both speakers use banana equally often: $p = 0.5$.
- Our text has two chocolates and one fudge. Speaker A uses chocolate more often than B.
- The full count pattern is more likely under A: 0.135 versus 0.030 under B.

These counts favour A under our simple model.